

TI-99 Console Tester Diagnostic

Public Beta 0.8 is a self-contained diagnostic BIOS for Geoff Trott's Console Tester hardware. It supports the TI-99/4, TI-99/4A, TI-99/4QI V2.2, and a TI-99/4A with missing or damaged GROMs. The LOAD switch can start the diagnostic without relying on the console's title screen or keyboard service.

IMPORTANT: THE FULL DIAGNOSTIC REQUIRES A HARDWARE MODIFICATION. Geoff Trott's original Console Tester is wired for a single 8 KiB EPROM at `>E000`. It must be modified to accept the additional 8 KiB diagnostic window at `>6000` before you install and use `HEXDIAG08.BIN`. Do not place the full 16 KiB diagnostic image into an unmodified board. Follow the illustrated [16 KiB hardware modification guide](#) before programming or installing the replacement device.

The fixed 8 KiB core performs automatic CPU, tester SRAM, scratchpad, VRAM, console ROM/GROM read-path, diagnostic ROM, and LED checks. The optional second 8 KiB window adds readable results and interactive keyboard, joystick, video, sound, VRAM, LED, and identification tests. The original TI-99/4 correctly skips the unsupported Graphics-II bitmap stage.

BETA - USE WITH CAUTION. The VRAM and memory tests are destructive. Power off before fitting or removing the tester, and do not run it with unsaved work. Verify every modification with the board unpowered before installing an EPROM.

Download and program Beta 0.8

The current release is in [release/beta-0.8](#).

- Program `HEXDIAG08.BIN` into a W27C512 for the tested 16 KiB modification.
- Use the repeated-core compatibility image first when bringing up an unmodified 8 KiB board or checking the original `>E000` decode.
- Verify the programmed device against `SHA256.TXT` before installation.

The 16 KiB programmer image is 65,536 bytes. It places the menu extension at `>6000->7FFF`, the fail-safe core at `>E000->FFFF`, and fills unused space with `>FF`. Do not program either raw 8 KiB component file as if it were the complete W27C512 image.

Build from source

Requirements:

- Windows PowerShell or PowerShell 7;
- Python 3; and
- [xdt99](#), normally checked out beside this repository.

Run `BUILD.CMD` to assemble the release. Run `VERIFY.CMD` to rebuild it and check the headers, memory layout, vectors, fixed tables, checksums, release aliases, and screen-text placement. Set `XAS99_PATH` if `xas99.py` is not in a sibling `xdt99` directory.

The active assembly order is listed in `FILES.TXT`. The sources use short, TI-friendly basenames and retain `.a99` as their host-side suffix:

1. `src/MEGAU7.a99` - relocated raster controller for tester U7 SRAM;
2. `src/HEXDIAG.a99` - fixed 8 KiB core at `>E000`; and
3. `src/HEXEXT.a99` - optional 8 KiB extension at `>6000`.

See the [source and build guide](#) before changing addresses or shared services. A distributable image must be made by `build.ps1`; a raw assembler invocation does not patch the zero-sum words or compose the W27C512 layout.

Documentation

- [Operator manual](#) - setup, startup, menus, lamp meanings, and troubleshooting
- [Hardware modification guide](#) - tested 16 KiB wiring and 32K, 40K, and later redesign options
- [Source and build guide](#) - source map, ABI, build, verification, and release procedure
- [Assembly memory map](#) - every checked AORG/XORG region, occupied range, and remaining headroom
- [Changelog](#) - functional, verification, and documentation changes in each public beta
- [Technical reference](#) - address map, tests, firmware profiles, and Beta 0.8 qualification
- [Lessons learned](#) - causes and guards for the major keyboard, VDP, sprite, timing, and packaging faults
- [Provenance and licensing](#) - original work, permitted third-party material, and attribution
- [Backlog](#) - investigations intentionally deferred beyond the frozen Beta 0.8 image

PDF copies are provided under `output/pdf/` for readers who prefer a printable manual.

Compatibility status

Beta 0.8 has been exercised on a stock TI-99/4A, a TI-99/4QI V2.2, a GROMless TI-99/4A using LOAD recovery, and a TI-99/4 fitted with a Pico9918. The `/4` joystick selectors are columns 5 and 6; the `/4A` family uses columns 6 and 7. The sprite page shows three visible groups at the original four-sprites-per-line limit. An F18A or Pico9918 configured for a higher limit may also show the yellow fifth entry; that is a configuration difference, not a failure.

Licensing and credit

Source code and build/test tooling are Apache-2.0. Original documentation and the illustrated modification material are CC BY 4.0. MegaDemo-derived material has its own permission and attribution boundary. The enhanced-VDP detector adapts MIT-licensed code supplied by Troy Schrapel; see [LICENSE.md](#), [NOTICE](#), and [docs/PROVENANCE.md](#).

Please credit **Jon Guidry (hexbus)** and point readers back to the canonical project: github.com/hexbus/ti99-console-tester.